

Active Inference Institute ~ 2023 Quarterly Roundtable #2

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Duration: 57:19 | **Uploaded:** Unknown **Transcript method:** auto_caption

hello everyone welcome to the active inference Institute quarterly Roundtable number two in 2023. it's June 30th 2023 and this is going to be our quarterly updates and ways that you can participate at The Institute and in the active ecosystem so welcome to the octave inference Institute we're a participatory online Institute that is communicating learning and practicing applied active inference you can find us at the links on this page and note that we'll be transitioning our emails over from activeinference gmail.com towards activeinference.institute domains this is recorded in an archive live stream so please provide us with feedback so that we can improve our work all backgrounds and perspectives are welcome and we'll be following video etiquette for live streams today in the quarterly meeting number two of the year we'll start at The Institute scale give some updates at The Institute level and then we'll look a little bit deeper into primarily education but also research and then I'll have some closing Thoughts by way of a presidential address about where we're at with active inference and if you're watching live please write comments and questions in the live chat and we will look to address those during okay so starting with the uh Institute scale some updates coming from The Institute here's one way that we can visualize the octave inference Institute in June we are within the active inference ecosystem as well as within other ecosystems and we have a nested structure Alex would you like to describe this oh yeah I think for these moments it's great to see this structure and understand what that is actually in place and working and acting and adapting and from very beginning we considered From perspective of systems thinking and From perspective of act different currents we always think about some kind of Mass state and hierarchical structure which can be builded and structured accordingly so having it right now like embodied it's great to see it totally agree at The Institute scale where these first updates are going to come from we have the officers the board of directors and the scientific Advisory Board we have two organizational units which Steward the development and support participation in projects and projects and learning groups are really where it happens on the

ground and so we're going to talk a lot more about these but this is the nesting and our own relationship with the ecosystem another way to see this is in terms of people and positions at The Institute so starting from the left we have casual engagement with our products and services into the niche so this might be the very video that you're listening to or watching right now it could be the transcript of this video that you're reading it might be the active inference ontology it might be the active block for a software package there's all these different open source products and services that we provide to the niche and one moment you will decide to participate in a project or a learning group and that can be more towards learning more towards the doing and leveraging your contributions at The Institute and some participants will choose to solidify their participation through volunteering or engaging with the internship program both of which are new and really excitingly developing well and then we haven't listed any names of the many of you who are in all of these different stages and you may want to play another role or another position in the Institute such as the scientific Advisory board board of directors or the officers so a lot of appreciation and thanks to everybody who's shown and not shown on this slide um we are really appreciative of how you all show up for the Institute Alex want to add anything on that um yeah even from this slide we can say that a lot of people are already involved in activity and provide great insights and different perspective and backgrounds and experience and yes a lot of people not showing here but we know when through our community platforms and so on so you can see this progress in terms of participation in activities and it's also great progress all right moving on we are excited that for our third applied active inference Symposium to be held in August we will have the theme of enacting ecosystems of shared intelligence active inference as a cognitive kernel we have a really Broad and Global co-organizer team which is very exciting so thanks to everybody who's in the co-organizing team for this Symposium we're just getting started on the organization so in the coming days and weeks we'll be providing more information on participating in a more formal call for presenters however if you're listening to this message now and you are really excited to do a presentation or a workshop or mentorship and be a part of the active inference ecosystem then please reach out to us Alex anything you want to add here but actually yes I think we can remember but it will be a third symposium and it's cool but we have but here earlier written for this type of activity and hopefully this event bring us and for people who will participate you know insights connections and ground for future developments in the research all right just uh summarize and update a few of the Institute level programs so volunteering at the Institute it's a way to engage it is just making it that much

easier to get involved with one or more projects and to start to get that recognition from The Institute understanding your contributions and also on your own resume your own work history your portfolio volunteering is really an incredible way to learn by doing if you want to apply or learn about organizational active inference or if you want to think about communication as active inference education in the context of active inference gripper and the gripper if you want a perspective Swap and see it multiple ways learning by doing on a project through volunteering we hope is a really unique opportunity and what you do will and does really matter so if you're taking notes on the margin of some paper or if you're developing lists that you felt like help your onboarding into active inference you can make a leveraged contribution by sharing that kind of information or if you're not quite sure what type of contribution or the impact you would have we can help find that kind of a niche for you the internship program is being a volunteer but wanting a little bit more structure around that we have about 20 currently active interns from all around the world different backgrounds levels of technical familiarity with the active inference ranging from truly some of the people who I believe are experts to people who just found out about active inference and wanted to use this program to accelerate their learning a lot also in the last weeks and months I would like to commend the two individuals who know who they are who have completed their internship this has really been a great honor to work with you both of you and to those who are still active I look forward to seeing those updates and continuing to work with you the internship consists of learning doing and also mentorship and facilitation so interns tend to participate in one or more course or textbook group or maybe help out with some live streams interns engage on projects and that is usually a minimum of two projects one is hosted at The Institute so some of the projects that we'll look at later in this round table but then also most of the interns have a personal or background research agenda so sometimes they make very small contributions to an Institute project but they're working really diligently on their dissertation or on their hobby project other times it's most of the contribution to the Institute project and then a little bit of self-guided research and we try to provide our interns with Support over multiple spatial and temporal scales foreign engagement in the broader digital cyber physical Niche we continue to grow on our different platforms and channels though these are just proxy measures in our Discord podcast Twitter and YouTube as well as mailing lists so we continue to communicate out to those who want to hear here's a big picture on Institute activities and you can find this information at the activities page that's uh listed here at the top we have type of project publication live streams such as the one happening right now and the active inference

Journal which picks up when the live stream hits stop and transcribes and curates publishes that transcript which we'll talk more about so live streams in journal are a great pair in the active block prints project we continue to build the soft and hard skills associated with generative modeling in the active inference tradition and so we use the complex adaptive systems modeling framework cadcad to implement different active inference generative models this is a really great place a lot of discussions about modeling Robotics and embodied JF claudger continues to offer his expertise and updates from this really incredible project where real Lego robots are implementing symbolic active inference so thanks Jeff for hosting this project the way you do in the active inference ontology we continue to refine and update the octave inference ontology which is the real language that we speak whether it happens to be deployed in English or visually or in a programming language or however the active inference ontology is how we're communicating and we develop that and offer it as a service to the niche in this project we have multiple ongoing learning groups which we'll talk more about soon but we have a textbook group with par at all the 2022 active inference textbook we have sanjiv namjoshi who's writing a really incredible work that will be a textbook published as fundamentals of active inference and we're Alpha Testing along with Sanjeev and right there alongside him so thanks Sanji for bringing this work in progress to The Institute and to to really sticking with the trouble this is going to be an incredible product and we have two courses one on physics as information processing and one on active inference for the social sciences which we'll address soon and also multiple times per week in our Discord we have open discussions for Education and Research so if this is all a lot or you're not sure about anything probably the best way to just drop in and find out would be to check out when and how we're meeting and come to the open discussions if you just want to like see what's up anything you want to add there Alex yeah good to see that projects are active and the mostly maybe for last period of time it's uh having that new courses which will describe later but heaven and feeling me it's growing it's provide great opportunities for second half of the year so true in the first quarterly Round Table we had no courses but as of now we can say that there are two courses that are really pushing the frontier of education and even research in Quantum information sciences and in social sciences so thanks again we'll get to those soon all right let's dive into a little more detail on eduactive so Alex would you like to first introduce and give a little thoughts on the active inference qualification system draft yeah we put it here like for now it's education iterative scale but future it's uh will be applied for all Institute structure and activities we should have some Cornerstone in terms of how we can measure

effectiveness of our educational activities and Educational Services in the beginning to understand courses and our materials how we change actually someone's mind and how deep this might need to be changed in some formal sense and we take we not started from Greenfield we just take one of a stable system of qualifications level from European and adapted for how we can see it in application to active inference and participations and projects and so on and for now we have a basic draft and we'll have discussions around these interested parties and different skills of Institute and finally have some some stable form location will apply to a different direction thanks I'll just add a few notes on this first just echoing your last point if you're interested in active inference professionalization education qualification systems even as a Critic we would really love to have you in the game so please participate early participate often and I'll just highlight a few pieces of this system draft that are really exciting first it begins at level zero which everybody de facto can have perhaps even upon or before birth then as the levels proceed active inference is always being connected with domains and systems of Interest so it's about yes and with active inference about connecting active inference to other fields and topics understanding a different domain's ontology robotics or Neuroscience or pure mathematics whatever it may be and working with active inference as you gain knowledge of the ontology and of the details of active inference but it's not just a knowledge-based qualification system we also are looking to integrate skills and even project responsibility and autonomy so just to give kind of an example of this levels one and two again currently very tentative drafts so for your engagement but in levels one and two people are able to act as basic Learners and collaborators and begin to get a grasp on what terms we're using in active inference and how do those Concepts relate to their field of interest and we see this every single day in the textbook group and in discussions someone says oh I'm working as a physical therapist and so in action that's like the person moving their arm that is where these levels are targeted level three begins to get into describing and understanding some of the technical details of what we do level four begins to approach the evaluation and recapitulation of technical models in active inference level five is related to comprehensive development and implementation as well as in the projects and autonomy exerting more supervision for self and others level six developing Advanced skills level seven engaging with very specialized problem-solving skills and level eight engaging in synthesis evaluation and critical problem solving so again can't emphasize enough it's just a draft and we are calling for your participation if this is something that you would be interested in using or engaging with but we think this is going to be a really important direction for the active inference ecosystem carry

on to the textbook group so in the active inference textbook group we are working on and through the par pazulo and friston 2022 textbook currently we have cohort three working on the second half of the book and cohort four on the first half of the book together these two cohorts compromise about 150 people not every person joins every meeting some people engage with the recordings asynchronously other people are in our shared epistemic niche just gleaning and contributing to this kind of collaborative stigmergic process as educators were focused on the continual delivery and deployment of questions and enrichment and discourse and just some ways that this plays out here are for example questions that people have asked live or before or after a synchronous meeting where we can connect the specific figures and the ontology terms to people's uncertainties and on the right is an example of an equation here's the variational free energy equation 2.5 and the way that we can use the ontology to unpack and connect some of the dots with these equations and in the last few weeks we've actually taken that even further and explored how to use language models through code AI in this process so here we have equations 2.1 and equations 2.2 they are being rendered in coda via typing in latex and then we are using language models within Coda to provide explanations contextualizations for different backgrounds and expertises and also translation into different languages describing the variables and the notation and the who what and why so this has been a pretty fun Avenue and a really cool example of how language models and education can be hand in hand and Empower our own learning by using all kinds of models to help us where frankly it can be quite challenging which is in understanding some of the analytical details of the framework anything you want to add Alex this is that type of technology which looks like magic and but and then but heaven in place ontology which is nicely integrated for Coda again to all projects it's really like imagine for this car which which is getting more and more big cool okay in May 2023 we began our first course at The Institute and that was physics as information processing being taught by one of the leading researchers in the area Chris fields and with a math PhD course assistant Andre Aguirre so Chris thank you for stepping up again from a scientific Advisory Board to offer this truly incredible and unique educational experience we've had massive response dozens of registrants who want to engage in the participatory discussions that happen in between your lectures the second discussion being tomorrow for those who want to join and 1500 views of the first lecture in a month there's many people who are following along the course syllabus is explicit and innovative we have a question and answer system so that people can ask questions asynchronously Chris then answers them and we have them posted again through this

interactive Coda interface and that's really supporting an incredible level of discourse and participation all transcripts are going to be curated and published along with the questions and answers and some other supplemental information so this has been a really great experience and something new over the last few months anything you want to add I'm tempted to just talk about how cool the Quantum information free energy principle is but perhaps now is not the moment um for me it's uh uh like historically for my all the ground stop change from beginning on this uh Century starting from I.T technology technologist and computer science and this notion of information processing it always was like in Engineering in engineering stands like a work in field and working domain and but I always had that need and feeling for first principles for physics in place and for now seeing with that latest developments on Quantum Quantum information Theory and how it's nicely connects to free energy and active inference it's really really great to have this Frontier curves uh in place in our activities and really really great thanks to police and Hunter and I believe that it's a kind of starting point maybe 10 years after we'll come back all together and discuss how this course started so great changes for different thank you Alex totally agree and also we are fresh off the presses happy to announce the second course that we'll be offering at The Institute which is constructing cultural landscapes active inference for the social sciences and this is going to be a multi-teacher massively participatory course Avail of Kairos research thank you for taking the initiative to suggest even this possibility and Ben and Mal Lorena thanks a lot for joining on for the journey on July 7th we will begin and it will have some lecture sessions as well as participatory discussion sections all sessions sessions are free they'll be live streamed and rewatchable the transcripts will be curated and published into a book at the end of the year and this is going to be a really great on-ramp and space for people with the humanities or social sciences background so we've heard people waiting for such a thing it was aval's Insight that such a course could be hosted at Kairos research and at the Active inference Institute so thanks to all the teachers here and just really looking forward to how you tackle and engage with this question of active inference in the social sciences anything you want to add uh yeah like this two different courses like from different to distant scales like from Quantum scale and social scale and possibly it's like two different ends of spectrum in which we can work on with our research and education but it's great that this step to social sciences because it brings active influence and it's formal and first place principled framework into the domain which haven't got it before and now it's a great beginning also and our scales between Quantum and social will be on us to work on next years yes and uh the kind of Aura Boris or horseshoe Theory

with Quantum social sciences onto the live streams so a lot to say of course a little bit meta since we're on a live stream right now but I'll just note a few things um first across our different series we now have 336 completed or planned live streams so most of those have already happened but there are several that are planned but we're over 300 completed and 336 as of now being planned and even just specifically looking at the next 30 days we have just such a breadth and depth it's quite exciting to watch let alone to enact we'll have the Chris Fields course a discussion section a lecture and then another discussion section we'll begin the active inference for the social sciences course with avell Ben having contributions we'll have a guest stream with professors from around the world researchers all these different topics being able to be slotted in with a really great way to platform and give visibility and catalyze discussion around these incredible researchers work so it's really exciting and it would be great not going to say it's specifically a personal request but if you choose to read it as such then great it would be awesome to have more assistance with participating and preparing and enacting live streams we have a role-based ways of planning and preparing and enacting live streams so whether you want to play a more technical or operational or logistical role if you want to be a conversational facilitator if you want to do it in a different language if you want to do a series there's so many ways that you can get involved but it would just be awesome to have more voices on the live stream and just be able to continue offering these kinds of resources at even faster rate with increased professionalization and I know that there's many people out there who would be incredible contributors to broadcast and or facilitate everyone is also welcome to self-nominate or suggest or introduce somebody who you think would fit well on a stream if it's like a perspective that we haven't heard from that would be really important there's a lot of background Technical and educational work on the dot zero videos when we do a paper discussion so if there's a paper that you want to have a a million x leveraged contribution on your preparation around if we discuss that in a live stream and you're able to help before with the dot zero you'll be a co-author on a research publication of the transcript and you'll have made an absolutely unique Irreplaceable contribution to active inference knowledge with what you did in your diligence in the dot zero so for people who hear that calling that's incredibly important we invite people to share the live streams on social media or with their colleagues however they find it relevant like if there's a topic or a person that has come up on our streams sharing it where relevant will help bring people in and also we have weekly live stream meetings so if you just want to find out like what we do and how we do it in the live streams then I would recommend that you join the regular meetings and if

you really want to join but that time never works for you then email and let us know that you're specifically interested we'll find a different time any other comments there Alex oh this moments we can see that oh but a lot of live stream activity is happening this participation of people not only with our own preparations about these people who just want to share this and so on and if we can keep this Rhythm to having different people with their interests and sharing ideas and yes if people will join to help us to do it more and more often Maybe it will be great thank you the active inference Journal picks up where live stream stops the active inference Journal is engaged in continuous deployment and development of speech to text transcription automated and manual editing curation and finally publication so just to give an example a live stream will occur and then we use speech to text onto a public GitHub repository everybody can edit the markdown file and then we're able to publish those transcripts so that they can be citable so that a conversation someone has can go onto their resume as a true citable publication and we can just facilitate the accessibility and the findability of these incredible conversations and moments that we have and also in the active inference Journal we're always developing the process from setup for those of all technical backgrounds how to get things working on your own local setup or on a cloud setup using free and open source software where possible the speech to text the text processing the editing translating into languages rendering into a diversity of formats publishing through different channels and then engaging in further analysis related to natural language processing and the active inference ontology all these fun things we can do so if you in an alternate Multiverse we're a librarian or an editor or a viewer or an advocate of the active inference Corpus bring it into this reality and make a huge contribution by cultivating our epistemic garden and our offerings because there's so much enrichment that can happen with past streams and with proactively letting the live stream project know and letting courses know how we can help them at the journal to develop these kinds of Publications that in new ways are expanding definitions of scholarship and research and service anything else to add as for this project I I believe it's like most technical in The Institute and maybe we can call it more probably like language processing project and maybe it is a good starting point for people who got who have good technical skills some programming skills which are related to language processing in general or some specific part of it and maybe it will be of interest for them to start from this project and later I get a better grasp on what is active influence is awesome totally agree it's like seeing the range of live streams is really one of the best ways to just jump into understanding the the waste and the topics that people speak we'll just briefly mention the

research work happening but this last several month period has been extremely focused and extremely productive on education we have the two new courses the qualification system all of these other curricular developments um so there's several ongoing research projects we have the active inference ontology as a research and development initiative active block prints as a software package for complex systems modeling using active inference Robotics and embodied with jf's symbolic active inference robots and then there's some project seeds that we always love to offer up to interns or to volunteers as something that they can like restart and take a lot of initiative something that's just planted but but barely there and that right now is the graphical interface with thinking really expansively about what it looks like to learn an apply active inference from a embodied perspective and cognitive agent modeling doing meta-analysis and synthesis across different cognitive modeling Frameworks so that we can really have a place to point when we describe how different cognitive phenomena speaking to a cognitive pattern language are implemented or addressed by active inference and other Frameworks and then there's the so many of you who engage in individual research so there's a lot of research happening in the space and we see education as a fundamental enablement for research as part of the institute's mission for Education and Research development so it's been good times with research and uh I'm really excited to see all the all the projects that people share during sessions start to develop and and flourish anything else you want to add about research Alex uh yeah maybe uh this direction was not so active last times but as we discussed for future directions that we will bring more attention and to more efforts to maybe formalize a bit per week as Institute service people around from ecosystems uh how we can support them on their ways and so on so it will be good times here too yeah agreed we want to provide a lot of research services so for people who who want to be on any side of that table Reach Out all right well the last section here is going to be a short active inference based presidential address I will be just speaking to some recent developments as well as past developments in the active inference space and just give a few thoughts on how they relate to the active inference Institute and what we're all doing here if you're watching live please write some questions or comments into the live chat and at the end of this short address we will review any other comments or questions from those who are here live give some last thoughts and then we'll be back into the work so I wanted to begin this address with a reference to the work that was before active inference Institute which was active inference lab but even before active inference lab there was the collaboration with Alex Yvonne Alexandra RJ and myself around active inference and behavior engineering for teams and organizations and this was in September 2020 only a few

years ago now more than two but less than three years ago and uh it's amazing how far things have come and at the same time how many of the topics that we were able to bring into this context of remote teams and still ring as relevance and they're just gestured to here we've gone far towards enacting and implementing and improving but still we can think about cognitive entities engaged in the Cyber physical niche with narrative and rhetoric structuring their understanding of how sensory data translate into action so for people who are interested in this topic we invite you one way or another to engage with us on this journey because we are only going to do it all together so just wanted to begin with that kind of callback to several years ago though so much has changed certainly for all of us in the times between that when we really find first principles approaches like active inference and when we really care about systems of interest and communities then the actions that we take we don't regret but I think we can go into a little bit more detail and reference some recent ecosystem that speak to how day after day week after week our system of Interest active inference learning and application really gifts or blesses us with new ways of thinking and new ways of acting so for example just within the last months the inner screen model of Consciousness was released as a preprint by Maxwell ramstead at all dealing with all these topics that we know and love and building upon the quantum free energy principle formalism which again it's easy to forget was not even in place two years ago discussed in live stream number 40. and weeks after publication we're able to enact a conversation hearing from voices all over the world bringing experts and those with different backgrounds onto the screen here the screen that you see is the classical information depository that enables us as Quantum cognitive agents to engage with each other and so all of these beautiful people on the Stream who with their regime of attention and their Quantum cognitive rotation brought to bear their attention and their language contribution through action we are able to inscribe that on classical digital artifacts like live streams and transcripts and then provide it to the niche for all of you to rotate that into your own space and context so again the spatial temporal and conceptual linkages are happening at a speed that is boggling and on the right is a paper from just this month officially by Thomas Parr at all cognitive effort and active inference and I have the image small because there's more details to go into in the paper however they used a well-known model of cognitive burden called a Stroop model which shows a color written out as a word and that word is shown in some color so the color red written in red the color red written in blue and the timing dynamics of people's reading behavior and recognition behavior is a proxy for cognitive effort and so this is just one more not even Brick in the Wall not even just one tool in the tool kit but like another lens or

layer or ability or capacity for active inference framework which adds these features affect memory anticipation all these different features are layered in like a symphony without developing bloat because the first principles kernel the cognitive firmware is not getting bloated it is staying at first principles and sometimes it feels like even diving down to zero with principles while Advanced research in the space is able to help us understand really sophisticated cognitive phenomena like cognitive effort now one can look back through the previous stream and what we discussed and how could cognitive effort help us support the accessibility rigor and applicability of active inference it's such a natural question and again week after week our Niche provides us with new ways of thinking and doing so it's so fun to be gripping and gripped within active inference I'd like to recognize and commend Jacob's excellent work just in April we published generalized notation notation for active inference models GNN generalized notation notation is a plain text format for conveying active inference generative models it's something like a dialect of markdown and it brings us closer and closer to the Triple Play The Triple Play is the idea that we should have integrity amongst multiple representations of active inference models that is what we see visualized graphically or tangibly what we hear and say in terms of the natural language use and with executable cognitive simulations so if someone looks at an organizational flowchart and there's three A's connected to two B's that is something that we can visualize describe and execute so that there isn't a front and a back room conversation rather there's an integrated conversation that allows for everybody to make contributions in the modalities and in the ways that they want and the interlingua not the only interlingua and by no means its final version is the plain text generalized notation notation we're having a lot of fun working with language models and visualizing this transforming it into executable simulations and so thanks again jaka for the really creative and dedicated work because this method is going to be a real core functionality for translating across visual elements our natural languages and the kinds of simulations that we want to run and I'll close with a recent publication with Carl Friston at all including myself and blue so two out of three officers on the paper not terrible and in this paper we used free energy principle and active inference to provide something like a measuring or an observability framework for Ecology Evolution and development or eco-evo Devo and the way that this was done is through the renormalization group of physics and so in this image on the left we can see it as like maybe a bunch of cells which each contain organelles and those cells make a tissue or the big circle might be the population of organisms or the ecosystem Community again recognizing the nested niche engagement within and across levels of spatial

and temporal modeling looking at just two layers and how they're connected we can think of that as a nested action perception cycle so here's like the behavior sensory um and the behavior of of a worm in this case happening at whatever time scale it's being modeled at and then at the slower or the deeper level of the hierarchical model we have the Bayesian learning like process of the extended genotype happening in a slower adaptive layer and so this should clearly connect to what we said at the very beginning of the stream about the Institute as part of a nested system here we have the ecosystem the space around here's the Institute this is what it really means to enter and to join not just to look into but to be on or in the Institute and then the kinds of dynamic partitions that were able to embody within the Institute through our Communications and and through our actions it's just so exciting to see everyone's incredible research the kinds of questions and developments that even if I were not an author even if I were not in the active inference field I would believe are at least the right things to be trying in terms of bringing theoretical integration in ecology Evolution and development at least these are the right research Avenues but then again to be able to act infer and serve through the Institute as a multi-scale evolutionary entity in and of itself this kind of real-time enactment of multiple scales and Quantum reference frames is something that is only arising through co-creation so in closing I send my deepest appreciation to all those who support and care and engage to those who are not sure I welcome you to the past the present and the future we respect and that's why we do what we do Alex where where do we go or what is next thank you very much for the talk and I'm really providing that so you know the reasons well from the beginning and about going now and congratulations on that publication on evolutionary work and it's a great opening Direction the least these direction will make a next step into techno Revolution also and we will use this Tech Revolution customer approach the develop Institute Canada consistent yes I'll just wait a few more seconds if anyone writes a question that we can address in the live chat let's look at some of these questions that we look at at the end of every live stream what would a good understanding enable well if we were able to really understand our own generative model as The Institute and as the active inference ecosystem it would support our sense making and decision making we'd be able to act and infer and serve better what are the unique predictions and implications that's a very interesting question I'm sure there are many ways to to address this and we we did have many future oriented Expressions during this discussion even today what are the next steps for free energy principle and active inference again do you have any thoughts on this one the middle question or the second question Alex and the unique predictions and next steps depends on some

specific field or domain or or all for ourselves we understand whether to conferences scale free approach and it just can be applied to any scale of research or any domain which uh researcher or engineer is working on can be described to his active internal language and stuff to informed from where using affordances to computational modeling or some other types of modeling yeah what are the goals of the research to embody our mission to support the active inference ecosystem to to be the best Institute we can be be the best in estimates within it what do you think maybe we should remove goals okay here let's change it to preferences yeah at least and considering in open-endedness way again we're okay we can speak about those ones it looks like it's not so useful concept yes sometimes when we're talking about the um specific research papers we talk about the goal of that paper but goal is not to be found in the active inference ontology we find ways of making do and thriving without needing to rest upon goals and reward and that's really exciting and that's something that a lot of people don't even realize that they're steeped in reward maximization and goal implementation until the open-endedness and the Vitality of active inference becomes understood what are you still curious about Ola you go first then I'll give the last thought as we close um I always remember our discussions about sales of the world curios in the beginning of it's like [Music] I believe it is a week a week each day each month what our appealth our Frontier are showing us so much a lot of different perspectives different connections real people in Communications and projects and so on so it's happiness very beautiful I'm curious about many things like that you said that we uh develop and hone that Curiosity every day and we we seek to to spark and be sparked when people come in from different backgrounds with their Curiosities it's inspiring to say the least and I'm really curious about who will have listened or read this far in the Stream and choose that moment to take an action to get involved with the Institute so there's only one way that my uncertainty is going to be reduced about that and I think if you know you know so thank you Alex thank you blue thank you to the officers and to all of the Institute see you around thank you thank you